Pre-Analysis Plan for Many Daughters Project

Team RT

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1. Introduction

This Pre-Analysis Plan (PAP) outlines our empirical strategy to test four causal hypotheses using the German Socio-Economic Panel (SOEP). As participants in the Many Daughters project, we are in the no-data group and will develop our plan without access to the data. Our focus is on the effect of having daughters on attitudes and behaviors related to gender equality and political preferences.

2. Hypotheses

- **H1**: Having daughters leads to stronger support for gender equality (attitudes).
- **H2**: Having daughters leads to more left-leaning political preferences.
- **H3**: Having daughters leads to more gender equality in household behavior.
- **H4**: The effect of having daughters on gender equality attitudes is stronger for men.

3. Data and Sample

Data Source: German Socio-Economic Panel (SOEP).
Inclusion Criteria:

- Respondents aged 18 and above.
- Respondents with at least one biological child.

4. Variable Construction

Treatment Variable:

- Indicator for having at least one daughter.

Outcome Variables and SOEP Sources:

- **Gender Equality Attitudes** (H1, H4): Based on items from SOEP core modules on gender roles.
- **Political Preferences** (H2): Political left-right self-placement scale.
- **Household Gender Equality** (H3): Division of domestic work, and employment status.

Control Variables:

- Age, education, household income, number of children, region dummies, year dummies.

5. Empirical Strategy

We estimate OLS regressions of the form:

$$Y_i = \alpha + \beta \cdot \text{HasDaughter}_i + \gamma \cdot X_i + \varepsilon_i$$

where Y_i is the outcome variable and X_i are covariates.

H4: Add interaction between respondent gender and having daughters.

6. Multiple Hypothesis Testing

We will adjust p-values using Bonferroni correction across H1–H4.

7. Robustness Checks

- Alternative treatment definitions (e.g., proportion of daughters).
- Subsamples by education or age.
- Additional controls.

8. Expected Outputs

- Tables with regression coefficients for H1–H4.
- Summary statistics of key variables.
- Graphs for visualizing interaction effects (H4).

9. Analysis Code Plan (Stata Template)

```
* Many Daughters - RT 01p8xt
clear all
set more off

cd "C:\Project\ManyDaughters"
log using logs/analysis_log.smcl, replace

* Main data source(s): SOEP-Core v39, BIOBIRTH

use "data/soep_sample_merged.dta", clear

* Create treatment
gen has_daughter = 0
replace has_daughter = 1 if kidsex01 == 2 | kidsex02 == 2 ///
            | kidsex03 == 2 | kidsex04 == 2 | kidsex05 == 2 ///
            | kidsex06 == 2 | kidsex07 == 2 | kidsex08 == 2 ///
            | kidsex09 == 2 | kidsex10 == 2

* Create outcomes

gen gender_eq_attitude = bjp_166_02
replace gender_eq_attitude = . if gender_eq_attitude < 0

gen political_orientation = bjp_171
replace political_orientation = . if political_orientation < 0
recode political_orientation (0=10) (1=9) (2=8) (3=7) (4=6) (5=5)
   (6=4) (7=3) (8=2) (9=1) (10=0)

gen household_equality = ks_cost_h

```

```
* Controls
gen age = ...
gen educ = ...
gen income = ...
gen num_kids = ...
gen male = (sex == 1)
gen region = ...

* H1
reg gender_eq_attitude has_daughter age educ income num_kids i.region i.year

* H2
reg political_orientation has_daughter age educ income num_kids i.region i.year

* H3
reg household_equality has_daughter age educ income num_kids i.region i.year

* H4
gen interaction = has_daughter * male
reg gender_eq_attitude has_daughter male interaction age educ income num_kids i.region i.year

log close
```

Listing 1: Stata Do-File Template

10. References

- SOEP Variable Search: `https://paneldata.org`
- SOEP Companion: `https://companion.soep.de`
- SOEP Survey Papers: `https://www.diw.de/en/diw_01.c.620272.en/publications/soep_survey_papers.html`